

6.4 Classwork: Systems of Equations — Markscheme

1. The diagram shows two lines, $y = 2x - 1$ and $y = -x + 5$.

- (a) Write down the gradient of each line. [2]

Gradient of $y = 2x - 1$: $m = 2$ (A1)

Gradient of $y = -x + 5$: $m = -1$ (A1)

- (b) Find the coordinates of the point of intersection by solving the system algebraically. [3]

$2x - 1 = -x + 5$ (M1)

$3x = 6 \Rightarrow x = 2$ (A1)

$y = 2(2) - 1 = 3$

Point of intersection: (2, 3) (A1)

- (c) Verify that the point satisfies both equations. [2]

Check $y = 2x - 1$: $y = 2(2) - 1 = 3$ ✓ (M1)

Check $y = -x + 5$: $y = -(2) + 5 = 3$ ✓ (A1)

2. The diagram shows two parabolas, $y = x^2 - 4x + 3$ and $y = -x^2 + 2x + 3$.

(a) Show that the x -coordinates of the points of intersection satisfy $2x^2 - 6x = 0$. [2]

$$x^2 - 4x + 3 = -x^2 + 2x + 3 \quad (\text{M1})$$

$$2x^2 - 6x = 0 \quad (\text{A1}) \quad (\text{AG})$$

(b) Hence find the x -coordinates of the points of intersection. [2]

$$2x(x - 3) = 0 \quad (\text{A1})$$

$$x = 0 \text{ or } x = 3 \quad (\text{A1})$$

(c) Find the coordinates of both points of intersection. [2]

$$x = 0: \quad y = 0 - 0 + 3 = 3 \quad \Rightarrow (0, 3) \quad (\text{A1})$$

$$x = 3: \quad y = 9 - 12 + 3 = 0 \quad \Rightarrow (3, 0) \quad (\text{A1})$$

(d) Find the coordinates of the vertex of $y = x^2 - 4x + 3$. [2]

$$x = \frac{-(-4)}{2(1)} = 2 \quad (\text{M1})$$

$$y = 4 - 8 + 3 = -1$$

$$\text{Vertex} = (2, -1) \quad (\text{A1})$$